

PROVABLE SCALING LAWS OF FEATURE EMERGENCE FROM LEARNING DYNAMICS OF GROKING

Yuandong Tian

Meta Superintelligence Labs (FAIR) & Independent Researcher
yuandong.tian@gmail.com

ABSTRACT

While the phenomenon of grokking, i.e., delayed generalization, has been studied extensively, it remains an open problem whether there is a mathematical framework that characterizes what kind of features will emerge, how and in which conditions it happens, and is still closely connected with the gradient dynamics of the training, for complex structured inputs. We propose a novel framework, named Li_2 , that captures three key stages for the grokking behavior of 2-layer nonlinear networks: (I) Lazy learning, (II) independent feature learning and (III) interactive feature learning. At the lazy learning stage, top layer overfits to random hidden representation and the model appears to memorize, and at the same time, the *back-propagated gradient* G_F from the top layer carries information about the target label, with a specific structure that enables each hidden node to learn their representation *independently*. Moreover, the independent dynamics follows exactly the *gradient ascent* of an energy function $\mathcal{E}$, and its local maxima are precisely the emerging features. We study whether these local-optima induced features are generalizable, their representation power, and how they change on sample size, in group arithmetic tasks. When hidden nodes start to interact in the later stage of learning, we provably show how G_F changes to focus on missing features that need to be learned. Our study sheds lights on roles played by key hyperparameters such as weight decay, learning rate and sample sizes in grokking, leads to provable scaling laws of feature emergence, memorization and generalization, and reveals the underlying cause why recent optimizers such as Muon can be effective, from the first principles of gradient dynamics. Our analysis can be extended to multi-layer architectures. The code is available¹.

1 INTRODUCTION

While modern deep models such as Transformers have achieved impressive empirical performance, it remains a mystery how such models acquire the knowledge during the training process. There have been ongoing arguments on whether the models can truly generalize beyond what it is trained on, or just memorize the dataset and performs poorly in out-of-distribution (OOD) data (??).

Modeling the memorization and generalization behaviors have been a goal of many works. One such behavior, known as *grokking* (?????), shows that the model initially overfits to the training set, and then suddenly generalizes to unseen test samples after continuous training. Many explanations exist, e.g., effective theory (??), efficiency of memorization and generalization circuits (?), Bayesian interpretation with weight decay as prior (?), etc. Most works focus on a direct explanation of its empirical behaviors, or leveraging property of very wide networks (???), but few explore the details of the grokking learning procedure by studying the gradient dynamics on the weights.

In this work, we propose a mathematical framework Li_2 that divides the grokking dynamics for 2-layer nonlinear networks into three major stages (Fig. 1). *Stage I: Lazy Learning*: when training begins, the top (output) layer learns first with random features from the hidden layer, and at the same time, the backpropagated gradient G_F to the hidden layer starts to carry useful signal about the target label (Proposition ??). *Stage II: Independent feature learning*: G_F drives the learning of hidden

¹<https://github.com/yuandong-tian/understanding/tree/main/ssl/real-dataset/cogo>

从 Grokking 学习动力学中推导特征涌现的可证明缩放定律

Yuandong Tian

Meta Superintelligence Labs (FAIR) & 独立研究员
yuandong.tian@gmail.com

Abstract

尽管延迟泛化的现象（即 grokking）已被广泛研究，但对于复杂结构化输入，是否存在一个数学框架能够刻画何种特征会涌现、如何以及在何种条件下发生，并且仍然与训练过程的梯度动力学紧密相连，这仍然是一个开放性问题。我们提出了一个名为 Li_2 的新框架，该框架捕捉了双层非线性网络 grokking 行为的三个关键阶段：(I) 惰性 学习，(II) 独立 特征学习，以及 (III) 交互 特征学习。在惰性学习阶段，顶层过拟合于随机的隐藏层表示，模型表现出记忆现象；与此同时，来自顶层的反向传播梯度 G_F 携带了关于目标标签的信息，并具有特定的结构，使得每个隐藏节点能够独立地学习其表示。此外，这种独立动力学精确地遵循能量函数 $\mathcal{E}$ 的梯度上升过程，其局部极大值点正是涌现的特征。我们研究了这些由局部最优诱导的特征是否可泛化、它们的表示能力，以及在群算术任务中它们如何随样本量变化。当隐藏节点在学习的后期阶段开始交互时，我们可证明地展示了 G_F 如何变化以聚焦于需要学习的缺失特征。我们的研究揭示了权重衰减、学习率和样本量等关键超参数在 grokking 中所扮演的角色，推导了特征涌现、记忆和泛化的可证明缩放定律，并从梯度动力学的基本原理揭示了为何近期优化器（如 Muon）可能有效的根本原因。我们的分析可以扩展到多层架构。代码已公开¹。

1 引言

尽管 Transformer 等现代深度模型已取得令人瞩目的实证性能，但这些模型如何在训练过程中获取知识仍是一个谜。关于模型是否能真正泛化到训练数据之外，还是仅仅记忆数据集并在分布外（OOD）数据上表现不佳，一直存在持续的争论 (??)。

对记忆与泛化行为进行建模一直是许多工作的目标。其中一种被称为 顿悟 (grokking) 的行为 (?????) 表明，模型最初会过拟合训练集，然后在持续训练后突然泛化到未见过的测试样本。对此存在多种解释，例如有效理论 (??)、记忆与泛化电路效率 (?)、以及将权重衰减视为先验的贝叶斯解释 (?) 等。大多数工作侧重于直接解释其经验行为，或利用极宽网络的性质 (???)，但很少有研究通过分析权重上的梯度动力学来深入探索顿悟学习过程的细节。

在本工作中，我们提出了一个数学框架 Li_2 ，将双层非线性网络的顿悟动力学划分为三个主要阶段（图 1）。阶段 I: 惰性学习 (Lazy Learning)：训练开始时，顶层（输出层）首先利用隐藏层的随机特征进行学习，同时，反向传播到隐藏层的梯度 G_F 开始携带关于目标标

¹<https://github.com/yuandong-tian/understanding/tree/main/ssl/real-dataset/cogo>

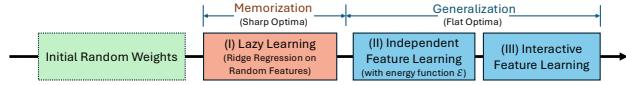


Figure 1: Overview of our framework Li_2 . Li_2 proposes three stages of the learning process, (I) Lazy learning, (II) independent feature learning and (III) interactive feature learning, to explain the dynamics of grokking that shows the network first memorizes then generalizes (see Fig. ?? for details). Our analysis goes beyond Neural Tangent Kernel (NTK) and mean field regime, and characterizes concretely how features emerge from gradient dynamics with the help of the energy function $\mathcal{E}$ (Thm. ??) and multiple key factors that affect the procedure. Specifically, we characterize the learned features as local maxima of $\mathcal{E}$ (Thm. ??) and the required sample size to maintain them (Thm. ??), establishing generalization/memorization scaling laws.

representations. In this stage, the backpropagated gradient of j -th neuron (node) only depends on its own activation, triggering independent feature learning for each node. *Stage III: Interactive feature learning:* When weights in the hidden layer get updated and are no longer independent, interactions across nodes adjust the learned feature to minimize the loss.

Each stage is studied in details. In *Stage I*, we show that the signal carried by the backpropagated gradient G_F either appears at the initial stage (Proposition ??) which depends on the top layer initialization, or the final stage (Lemma ??) which depends on the weight decay η . In *Stage II*, we find that the dynamics that governs independent feature learning exactly follows the *gradient ascent* of an energy function $\mathcal{E}$ (Thm. ??), which corresponds to nonlinear canonical-correlation analysis (CCA) between the input X and target Y . In group arithmetic tasks, we completely characterize the local maxima structure of $\mathcal{E}$ (Thm. ??), each representing one learned feature. We further show that the amount of training samples determine whether these features/local optima remain stable (Thm. ??), whether they are generalizable or not, and thus determine the scaling laws of generalization and memorization. In *Stage III*, we provably show that there is a push for diversity between hidden nodes with similar activations (Thm. ??), when a subset of features are learned, backpropagated gradient G_F changes to push the model to focus on missing features to minimize the loss (Thm. ??), and optimizers like Muon (?) further improves the diversity (Thm. ??). Experiments on graph arithmetic tasks support our claims, e.g., the proved scaling laws about the generalization/memorization boundary (Thm. ??) fits well with the experiments (Fig. ??).

Comparison with existing grokking frameworks. Our framework provides a theoretical foundation from first principles (i.e., gradient dynamics) that explains the empirical hypothesis ? that “generalization circuits C_{gen} is more efficient but learn slower than memorization circuits C_{mem} ”. Specifically, we show that the data distribution determines the optimization landscape, which in turn governs which local optima the weights converge into, which lead to the behavior of memorization or generalization. We also show that the initial memorization, or lazy learning (Stage I), has to happen before feature learning (Stage II-III), since the former provides meaningful backpropagated gradient G_F for the latter to start developing. In comparison, (?) also provides a three stage framework of grokking, but mostly from empirical observations. Using our framework, we provide a systematic framework to explain when and why the grokking happens that are consistent with many empirical observations (Sec. ??).

Our study goes beyond Neural Tangent Kernel (NTK) (??) and mean field regime (?), in which the hidden weights do not update substantially from initializations. Our framework demonstrates concretely how features emerge from gradient dynamics and multiple key factors that affect the procedure.

Summary. Our framework Li_2 splits the network training process into three stages (Lazy, independent, interactive), and provides insights into grokking dynamics and feature learning.

- *Explanation of Grokking behavior.* At the beginning of grokking, the lazy learning corresponds to memorization, in which the top layer finds a (temporary) solution to explain the target with the random features. Only after that, the backpropagated gradient G_F becomes meaningful, and causes the hidden layer to learn generalizable, emerging features. Grokking is accelerated with regularization.

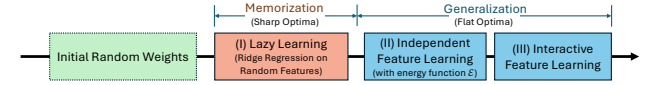


图 1: 我们的框架 Li_2 概述。 Li_2 提出了学习过程的三个阶段: (I) 懒惰学习、(II) 独立特征学习以及 (III) 交互特征学习, 用以解释网络先记忆后泛化的“顿悟”动态 (详见 Fig. ??)。我们的分析超越了神经正切核 (NTK) 和平均场机制, 并借助能量函数 $\mathcal{E}$ (Thm. ??) 以及影响过程的多个关键因素, 具体刻画了特征如何从梯度动态中涌现。具体而言, 我们将学习到的特征刻画为 $\mathcal{E}$ 的局部极大值 (Thm. ??), 并给出了维持这些特征所需的样本量 (Thm. ??), 从而建立了泛化/记忆的标准律。

签的有用信号 (命题 ??)。阶段 II: **独立特征学习** (Independent feature learning): G_F 驱动隐藏层表示的学习。在此阶段, 第 j 个神经元 (节点) 的反向传播梯度仅取决于其自身的激活, 从而触发每个节点的独立特征学习。阶段 III: **交互式特征学习** (Interactive feature learning): 当隐藏层权重被更新且不再独立时, 节点间的相互作用会调整已学习的特征以最小化损失。

我们对每个阶段进行了详细研究。在阶段 I 中, 我们表明反向传播梯度 G_F 所携带的信号要么出现在初始阶段 (Proposition ??), 这取决于顶层的初始化; 要么出现在最终阶段 (Lemma ??), 这取决于权重衰减 η 。在阶段 II 中, 我们发现支配独立特征学习的动态过程精确地遵循能量函数 $\mathcal{E}$ 的梯度上升 (Thm. ??), 这对应于输入 X 与目标 Y 之间的非线性典型相关分析 (CCA)。在群算术任务中, 我们完整刻画了 $\mathcal{E}$ 的局部极大值结构 (Thm. ??), 每个极大值代表一个学习到的特征。我们进一步证明, 训练样本的数量决定了这些特征/局部最优是否保持稳定 (Thm. ??), 决定了它们是否可泛化, 从而决定了泛化与记忆的标准律。在阶段 III 中, 我们可证明地表明, 当一部分特征被学习后, 具有相似激活的隐藏节点之间存在一种促进多样性的推力 (Thm. ??), 反向传播梯度 G_F 会发生变化, 推动模型关注缺失的特征以最小化损失 (Thm. ??), 而像 Muon (?) 这样的优化器则进一步提升了多样性 (Thm. ??)。在图算术任务上的实验支持了我们的主张, 例如, 所证明的关于泛化/记忆边界的标准律 (Thm. ??) 与实验数据吻合良好 (Fig. ??)。与现有顿悟框架的比较。我们的框架从第一性原理 (即梯度动力学) 提供了理论基础, 解释了经验假设 ? : “泛化电路 C_{gen} 比记忆电路 C_{mem} 更高效但学习更慢”。具体而言, 我们表明数据分布决定了优化景观, 而优化景观进而控制权收敛到哪个局部最优解, 从而导致记忆或泛化的行为。我们还表明, 初始的记忆化, 或称惰性学习 (阶段 I), 必须发生在特征学习 (阶段 II-III) 之前, 因为前者为后者开始发展提供了有意义的反向传播梯度 G_F 。相比之下, (?) 也提供了一个三阶段的顿悟框架, 但主要基于经验观察。利用我们的框架, 我们提供了一个系统性的框架来解释顿悟何时以及为何发生, 这与许多经验观察结果一致 (第 ?? 节)。

我们的研究超越了神经正切核 (NTK) (??) 和平均场机制 (?), 在这些机制中隐藏权重从初始化开始没有发生实质性更新。我们的框架具体展示了特征如何从梯度动力学中涌现, 以及影响该过程的多个关键因素。

摘要。 我们的框架 Li_2 将网络训练过程划分为三个阶段 (Lazy、independent、interactive), 并提供了对顿悟动态和特征学习的见解。

- **顿悟行为的解释。** 在顿悟开始时, 惰性学习对应于记忆, 其中顶层利用随机特征找到一个 (临时的) 解决方案来解释目标。此后, 反向传播的梯度 G_F 才变得有意义, 并导致隐藏层学习可泛化的、涌现的特征。正则化会加速顿悟。

- *Emerging features* are the local maxima of an energy function $\mathcal{E}$ that governs the independent stage. These features are more efficient in label prediction than simple memorization.
- *Data* governs the landscape of $\mathcal{E}$. Sufficient training data maintain the shape of those generalizable local maxima, while insufficient data lead to non-generalizable local maxima.
- *Scaling Laws of Feature Emergence, Generalization and Memorization* can be derived by inspecting how the landscape changes with the data distribution.
- **Independent feature learning.** $\Pr[\mathbf{w} \rightarrow \zeta_l] = p_l := \mu_l^a / \sum_l \mu_l^a$. Then the expected #nodes to get all local maxima is $T_0 \geq \max \left(1 / \min_l p_l, \sum_{l=1}^L 1/l \right)$.
- **Muon guiding.** If we use Muon optimizer to optimize K nodes sequentially, then the expected #nodes to get all local maxima is $T_a = 2^{-a}T_0 + (1 - 2^{-a})L$. For large a , $T_a \sim L$.

(A) If ϕ is *nondecreasing* on $(0, \sqrt{2}]$, then the unique maximizer is the *memorization* (peaked) solution

$$s_{g^*}^* = \sqrt{2}, \quad s_{g \neq g^*}^* = 0,$$

realized by $u = \frac{1}{\sqrt{2}}e_{g^*}$, $v = \frac{1}{\sqrt{2}}e_{(g^*)^{-1}h}$.

(B) If ϕ is *strictly decreasing* on $(0, \infty)$, then the unique maximizer *spreads* and is given by

$$s_g^* = \phi^{-1} \left(\frac{2\lambda}{p_g} \right) \quad (\text{for all } g \text{ with } 2\lambda/p_g < \phi(0^+)),$$

and $s_g^* = 0$ for any g with $2\lambda/p_g \geq \phi(0^+)$ (if $\phi(0^+) < \infty$). The multiplier $\lambda > 0$ is uniquely determined by the budget $\sum_g (s_g^*)^2 = 2$. In particular, if $\phi(0^+) = \infty$ (e.g., ReLU on $[0, \infty)$: $\phi(x) = 1/x$; SiLU: $\phi(x) = \frac{\text{sigmoid}(x)}{x} + \text{sigmoid}(x)(1 - \text{sigmoid}(x))$), then all coordinates are strictly positive and

$$p_i > p_j \implies s_i^* > s_j^* > 0.$$

- For power activations $\sigma(x) = x^q$ ($q \geq 2$) have $\phi(x) = q x^{q-2}$ nondecreasing; Thm. ??(A) gives memorization. In all these cases, the peaked solution is realized by even split $u = \frac{1}{\sqrt{2}}e_{g^*}$, $v = \frac{1}{\sqrt{2}}e_{(g^*)^{-1}h}$; any profile s^* can be realized with, e.g., $u_g = v_{g^{-1}h} = s_g^*/2$.
- ReLU on $[0, \infty)$: $\sigma(x) = x$, $\phi(x) = 1/x$ strictly decreasing; Thm. ??(B) yields $s^* \propto p$.
- SiLU/Swish/Tanh/Sigmoid: ϕ strictly decreasing with $\phi(0^+) = \infty$; Thm. ??(B) gives a strictly ordered spread $s_g^* = \phi^{-1}(2\lambda/p_g)$.

(A1) $\tilde{F}$ has full column rank K , and

(A2) $\text{col}(\tilde{Y}) \subseteq \text{col}(\tilde{F})$, i.e. there exists $V^* \in^{K \times M}$ such that $\tilde{Y} = \tilde{F}V^*$.

(A3) Small and independent random initialization on entries of $V(0)$, with mean zero and variance α^2 , where $0 < \alpha \ll 1$, and thus $\|V(0)\|_F = O(\alpha\sqrt{KM})$ with high probability.

(A4) Zero-mean centering: $\mathbf{1}^\top \tilde{F} = 0$ and $\mathbf{1}^\top \tilde{Y} = 0$.

- the *small-time expansion* of $G_{\tilde{F}}(t)$ and show that the leading term is $\tilde{Y}\tilde{Y}^\top \tilde{F}$; and
- the *long-time decay* behavior of $G_{\tilde{F}}(t)$, for both $\eta = 0$ and $\eta > 0$.

1.1 THE DYNAMICS OF $G_{\tilde{F}}$ AT INITIAL TIME STAMPS

1.1.1 SMALL-TIME EXPANSION AND LEADING TERM

Write the Taylor expansions at $t = 0$ as

$$V(t) = V_0 + tV_1 + O(t^2), \quad E(t) = E_0 + tE_1 + O(t^2),$$

where $V_0 := V(0)$ and $E_0 := \tilde{Y} - \tilde{F}V_0$. From (??),

$$V_1 = \frac{dV}{dt} \Big|_{t=0} = -AV_0 + B = -(\tilde{F}^\top \tilde{F} + \eta I_K)V_0 + \tilde{F}^\top \tilde{Y}. \quad (1)$$

- *涌现特征* 是控制独立阶段的能量函数 $\mathcal{E}$ 的局部最大值。这些特征在标签预测上比简单的记忆更高效。
- *数据* 控制着 $\mathcal{E}$ 的景观。充足的训练数据维持这些可泛化局部最大值的形状，而数据不足则会导致不可泛化的局部最大值。
- *特征涌现、泛化与记忆的缩放定律* 可以通过检查景观如何随数据分布变化而推导出来。
- **独立特征学习.** $\Pr[\mathbf{w} \rightarrow \zeta_l] = p_l := \mu_l^a / \sum_l \mu_l^a$ 。那么获得所有局部最大值的期望节点数为 $T_0 \geq \max \left(1 / \min_l p_l, \sum_{l=1}^L 1/l \right)$ 。
- **Muon 引导.** 如果我们使用 Muon 优化器顺序优化 K 个节点，那么获得所有局部最大值的期望节点数为 $T_a = 2^{-a}T_0 + (1 - 2^{-a})L$ 。对于大的 a , $T_a \sim L$ 。

(A) 如果 ϕ 在 $(0, \sqrt{2}]$ 上 非递减，则唯一的极大值是 记忆（峰值）解

$$s_{g^*}^* = \sqrt{2}, \quad s_{g \neq g^*}^* = 0,$$

由 $u = \frac{1}{\sqrt{2}}e_{g^*}$, $v = \frac{1}{\sqrt{2}}e_{(g^*)^{-1}h}$ 实现。

(B) 如果 ϕ 在 $(0, \infty)$ 上 严格递减，则唯一的极大值 分散，并由下式给出

$$s_g^* = \phi^{-1} \left(\frac{2\lambda}{p_g} \right) \quad (\text{对所有满足 } 2\lambda/p_g < \phi(0^+) \text{ 的 } g),$$

且对于任何满足 $2\lambda/p_g \geq \phi(0^+)$ 的 g (如果 $\phi(0^+) < \infty$)，有 $s_g^* = 0$ 。乘子 $\lambda > 0$ 由预算 $\sum_g (s_g^*)^2 = 2$ 唯一确定。特别地，如果 $\phi(0^+) = \infty$ (例如， $[0, \infty)$ 上的 ReLU: $\phi(x) = 1/x$; SiLU: $\phi(x) = \frac{\text{sigmoid}(x)}{x} + \text{sigmoid}(x)(1 - \text{sigmoid}(x))$)，则所有坐标都严格为正，且

$$p_i > p_j \implies s_i^* > s_j^* > 0.$$

- 对于幂激活函数 $\sigma(x) = x^q$ ($q \geq 2$)，有 $\phi(x) = q x^{q-2}$ 非递减；定理 ??(A) 给出记忆解。在所有情况下，峰值解由均等分割实现 $u = \frac{1}{\sqrt{2}}e_{g^*}$, $v = \frac{1}{\sqrt{2}}e_{(g^*)^{-1}h}$ ；任何分布 s^* 都可以用例如 $u_g = v_{g^{-1}h} = s_g^*/2$ 实现。
- $[0, \infty)$ 上的 ReLU: $\sigma(x) = x$, $\phi(x) = 1/x$ 严格递减；定理 ??(B) 得出 $s^* \propto p$ 。
- SiLU/Swish/Tanh/Sigmoid: ϕ 严格递减且 $\phi(0^+) = \infty$ ；定理 ??(B) 给出严格有序的分散解 $s_g^* = \phi^{-1}(2\lambda/p_g)$ 。

(A1) $\tilde{F}$ 具有满列秩 K ，且

(A2) $\text{col}(\tilde{Y}) \subseteq \text{col}(\tilde{F})$ ，即存在 $V^* \in^{K \times M}$ 使得 $\tilde{Y} = \tilde{F}V^*$ 。

(A3) 对 $V(0)$ 的条目进行小且独立的随机初始化，均值为零，方差为 α^2 ，其中 $0 < \alpha \ll 1$ ，因此以高概率有 $\|V(0)\|_F = O(\alpha\sqrt{KM})$ 。

(A4) 零均值中心化: $\mathbf{1}^\top \tilde{F} = 0$ 且 $\mathbf{1}^\top \tilde{Y} = 0$ 。

- 小时间展开 $G_{\tilde{F}}(t)$ 并证明其主导项为 $\tilde{Y}\tilde{Y}^\top \tilde{F}$ ；以及
- 长时间衰减行为 $G_{\tilde{F}}(t)$ ，针对 $\eta = 0$ 和 $\eta > 0$ 两种情况。

1.1 $G_{\tilde{F}}$ 在初始时间戳处的动力学

1.1.1 小时间展开与主导项

在 $t = 0$ 处写出泰勒展开式：

$$V(t) = V_0 + tV_1 + O(t^2), \quad E(t) = E_0 + tE_1 + O(t^2),$$

Differentiating $E(t) = \tilde{Y} - \tilde{F}V(t)$ gives

$$E_1 = \frac{dE}{dt}\Big|_{t=0} = -\tilde{F}V_1 = \tilde{F}(\tilde{F}^\top \tilde{F} + \eta I_K)V_0 - \tilde{F}\tilde{F}^\top \tilde{Y}. \quad (2)$$

Now expand $G_{\tilde{F}}(t)$:

$$G_{\tilde{F}}(t) = E(t)V(t)^\top = (E_0 + tE_1)(V_0 + tV_1)^\top + O(t^2) = E_0V_0^\top + t(E_0V_1^\top + E_1V_0^\top) + O(t^2).$$

Using $E_0 = \tilde{Y} - \tilde{F}V_0$ and V_1 from (1),

$$\begin{aligned} E_0V_1^\top &= (\tilde{Y} - \tilde{F}V_0)(-V_0^\top(\tilde{F}^\top \tilde{F} + \eta I_K) + \tilde{Y}^\top \tilde{F}) \\ &= \tilde{Y}\tilde{Y}^\top \tilde{F} - \tilde{F}V_0\tilde{Y}^\top \tilde{F} - \tilde{Y}V_0^\top(\tilde{F}^\top \tilde{F} + \eta I_K) + \tilde{F}V_0V_0^\top(\tilde{F}^\top \tilde{F} + \eta I_K). \end{aligned}$$

Every term except $\tilde{Y}\tilde{Y}^\top \tilde{F}$ contains (at least one factor of) V_0 , hence is $O(\alpha)$ in Frobenius norm. Moreover, $E_1V_0^\top$ also contains V_0 :

$$E_1V_0^\top = \tilde{F}(\tilde{F}^\top \tilde{F} + \eta I_K)V_0V_0^\top - \tilde{F}\tilde{F}^\top \tilde{Y}V_0^\top,$$

so $\|E_1V_0^\top\|_F = O(\alpha)$ as well. We therefore obtain the small-time expansion

$$G_{\tilde{F}}(t) = \underbrace{\tilde{Y}V_0^\top}_{O(\alpha)} + t\tilde{Y}\tilde{Y}^\top \tilde{F} + tR_1(V_0) + O(t^2), \quad (3)$$

where $R_1(V_0)$ collects all order- t terms that contain V_0 and thus satisfy $\|R_1(V_0)\|_F = O(\alpha)$.

1.1.2 WHY $\tilde{Y}\tilde{Y}^\top \tilde{F}$ IS THE LEADING TERM

We now compare the deterministic term $\tilde{Y}\tilde{Y}^\top \tilde{F}$ to the V_0 -dependent terms using norm inequalities.

Lemma 1 (Lower bound on $\|\tilde{Y}\tilde{Y}^\top \tilde{F}\|_F$). *Let $\tilde{F}$ have full column rank and $\tilde{Y}$ be nonzero. Then*

$$\|\tilde{Y}\tilde{Y}^\top \tilde{F}\|_F \geq \sigma_{\min}(\tilde{F}) \|\tilde{Y}\tilde{Y}^\top\|_F > 0,$$

where $\sigma_{\min}(\tilde{F})$ is the smallest singular value of $\tilde{F}$.

Proof. For any matrices A, B , $\|AB\|_F^2 = \text{tr}(BB^\top A^\top A)$. Take $A = \tilde{Y}\tilde{Y}^\top$, $B = \tilde{F}$. Since $BB^\top$ is PSD with eigenvalues bounded below by $\sigma_{\min}(\tilde{F})^2$,

$$\|AB\|_F^2 = \text{tr}(BB^\top A^\top A) \geq \sigma_{\min}(\tilde{F})^2 \text{tr}(A^\top A) = \sigma_{\min}(\tilde{F})^2 \|A\|_F^2.$$

Taking square roots gives the result. $\square$

Next, bound the V_0 -dependent part. For concreteness, consider the term $\tilde{F}\tilde{F}^\top \tilde{Y}V_0^\top$ (other mixed terms are bounded similarly). Using $\|AB\|_F \leq \|A\|_F \|B\|_F$,

$$\|\tilde{F}\tilde{F}^\top \tilde{Y}V_0^\top\|_F \leq \|\tilde{F}\tilde{F}^\top \tilde{Y}\|_F \|V_0\|_F.$$

Under the iid initialization with variance α^2 , $\|V_0\|_F = O(\alpha\sqrt{KM})$, hence

$$\|\tilde{F}\tilde{F}^\top \tilde{Y}V_0^\top\|_F = O(\alpha).$$

The same argument applies to all other V_0 -dependent order- t terms in $R_1(V_0)$. Combining Lemma 1 with these upper bounds yields

$$\frac{\|R_1(V_0)\|_F}{\|\tilde{Y}\tilde{Y}^\top \tilde{F}\|_F} \leq C(\tilde{F}, \tilde{Y}, K, M) \alpha$$

for some constant C independent of α . Thus, in the limit $\alpha \rightarrow 0$ (small random initialization), the term $\tilde{Y}\tilde{Y}^\top \tilde{F}$ is the unique leading contribution at order t .

Proposition 1 (Small-time leading term of $G_{\tilde{F}}$). *Under assumptions (A1)–(A2) and small random initialization with scale $\alpha \ll 1$,*

$$G_{\tilde{F}}(t) = \tilde{Y}V_0^\top + t\tilde{Y}\tilde{Y}^\top \tilde{F} + O(t\alpha + t^2)$$

in Frobenius norm. In particular, as $\alpha \rightarrow 0$,

$$\frac{G_{\tilde{F}}(t) - \tilde{Y}V_0^\top}{t} \rightarrow \tilde{Y}\tilde{Y}^\top \tilde{F}, \quad \text{and} \quad \|G_{\tilde{F}}(t)\|_F \sim t \|\tilde{Y}\tilde{Y}^\top \tilde{F}\|_F$$

for fixed small t , independently of whether $\eta = 0$ or $\eta > 0$.

其中 $V_0 := V(0)$ 且 $E_0 := \tilde{Y} - \tilde{F}V_0$ 。由 (??),

$$V_1 = \frac{dV}{dt}\Big|_{t=0} = -AV_0 + B = -(\tilde{F}^\top \tilde{F} + \eta I_K)V_0 + \tilde{F}^\top \tilde{Y}. \quad (1)$$

对 $E(t) = \tilde{Y} - \tilde{F}V(t)$ 求导得

$$E_1 = \frac{dE}{dt}\Big|_{t=0} = -\tilde{F}V_1 = \tilde{F}(\tilde{F}^\top \tilde{F} + \eta I_K)V_0 - \tilde{F}\tilde{F}^\top \tilde{Y}. \quad (2)$$

现在展开 $G_{\tilde{F}}(t)$:

$$G_{\tilde{F}}(t) = E(t)V(t)^\top = (E_0 + tE_1)(V_0 + tV_1)^\top + O(t^2) = E_0V_0^\top + t(E_0V_1^\top + E_1V_0^\top) + O(t^2).$$

利用 $E_0 = \tilde{Y} - \tilde{F}V_0$ 和来自 (1) 的 V_1 ,

$$\begin{aligned} E_0V_1^\top &= (\tilde{Y} - \tilde{F}V_0)(-V_0^\top(\tilde{F}^\top \tilde{F} + \eta I_K) + \tilde{Y}^\top \tilde{F}) \\ &= \tilde{Y}\tilde{Y}^\top \tilde{F} - \tilde{F}V_0\tilde{Y}^\top \tilde{F} - \tilde{Y}V_0^\top(\tilde{F}^\top \tilde{F} + \eta I_K) + \tilde{F}V_0V_0^\top(\tilde{F}^\top \tilde{F} + \eta I_K). \end{aligned}$$

除 $\tilde{Y}\tilde{Y}^\top \tilde{F}$ 外, 每一项都包含 (至少一个因子) V_0 , 因此在 Frobenius 范数下是 $O(\alpha)$ 。此外, $E_1V_0^\top$ 也包含 V_0 :

$$E_1V_0^\top = \tilde{F}(\tilde{F}^\top \tilde{F} + \eta I_K)V_0V_0^\top - \tilde{F}\tilde{F}^\top \tilde{Y}V_0^\top,$$

所以 $\|E_1V_0^\top\|_F = O(\alpha)$ 同样成立。因此我们得到小时间展开式:

$$G_{\tilde{F}}(t) = \underbrace{\tilde{Y}V_0^\top}_{O(\alpha)} + t\tilde{Y}\tilde{Y}^\top \tilde{F} + tR_1(V_0) + O(t^2), \quad (3)$$

其中 $R_1(V_0)$ 收集了所有包含 V_0 的一阶 t 项, 因此满足 $\|R_1(V_0)\|_F = O(\alpha)$ 。

1.1.2 为何 $\tilde{Y}\tilde{Y}^\top \tilde{F}$ 是主导项

我们现在使用范数不等式比较确定性项 $\tilde{Y}\tilde{Y}^\top \tilde{F}$ 与依赖于 V_0 的项。

Lemma 1 ($\|\tilde{Y}\tilde{Y}^\top \tilde{F}\|_F$ 的下界). 设 $\tilde{F}$ 列满秩且 $\tilde{Y}$ 非零。则

$$\|\tilde{Y}\tilde{Y}^\top \tilde{F}\|_F \geq \sigma_{\min}(\tilde{F}) \|\tilde{Y}\tilde{Y}^\top\|_F > 0,$$

其中 $\sigma_{\min}(\tilde{F})$ 是 $\tilde{F}$ 的最小奇异值。

证明. 对于任意矩阵 A, B , $\|AB\|_F^2 = \text{tr}(BB^\top A^\top A)$ 。取 $A = \tilde{Y}\tilde{Y}^\top$, $B = \tilde{F}$ 。由于 $BB^\top$ 是半正定的, 其特征值下界为 $\sigma_{\min}(\tilde{F})^2$,

$$\|AB\|_F^2 = \text{tr}(BB^\top A^\top A) \geq \sigma_{\min}(\tilde{F})^2 \text{tr}(A^\top A) = \sigma_{\min}(\tilde{F})^2 \|A\|_F^2.$$

取平方根即得结果。 $\square$

接下来, 界定依赖于 V_0 的部分。为具体起见, 考虑项 $\tilde{F}\tilde{F}^\top \tilde{Y}V_0^\top$ (其他混合项可用类似方式界定)。利用 $\|AB\|_F \leq \|A\|_F \|B\|_F$,

$$\|\tilde{F}\tilde{F}^\top \tilde{Y}V_0^\top\|_F \leq \|\tilde{F}\tilde{F}^\top \tilde{Y}\|_F \|V_0\|_F.$$

在方差为 α^2 的独立同分布初始化下, $\|V_0\|_F = O(\alpha\sqrt{KM})$, 因此

$$\|\tilde{F}\tilde{F}^\top \tilde{Y}V_0^\top\|_F = O(\alpha).$$

同样的论证适用于 $R_1(V_0)$ 中所有其他依赖于 V_0 的 t 阶项。将引理 1 与这些上界结合, 得到

$$\frac{\|R_1(V_0)\|_F}{\|\tilde{Y}\tilde{Y}^\top \tilde{F}\|_F} \leq C(\tilde{F}, \tilde{Y}, K, M) \alpha$$

其中常数 C 与 α 无关。因此, 在极限 $\alpha \rightarrow 0$ (小随机初始化) 下, 项 $\tilde{Y}\tilde{Y}^\top \tilde{F}$ 是 t 阶上唯一的主导贡献。

Remark on the role of η . The weight decay parameter η only appears in products involving V_0 , and hence all η -dependent order- t contributions are also $O(\alpha)$ in norm. Therefore the leading deterministic term $\tilde{Y}\tilde{Y}^\top \tilde{F}$ is the same for both $\eta = 0$ and $\eta > 0$.

1.2 LONG-TIME DECAY OF $G_{\tilde{F}}$

We now analyze the behavior of $G_{\tilde{F}}(t)$ as $t \rightarrow \infty$, again for both $\eta = 0$ and $\eta > 0$.

1.2.1 GENERAL SOLUTION OF THE GRADIENT FLOW

From (??), the gradient flow is a linear ODE with constant coefficients. The unique fixed point V^* satisfies

$$AV^* = B \quad \Rightarrow \quad V^* = A^{-1}B.$$

Define $\Delta V(t) := V(t) - V^*$. Then

$$\frac{d}{dt}\Delta V(t) = -A\Delta V(t), \quad \Delta V(t) = e^{-At}\Delta V(0),$$

and hence

$$V(t) = e^{-At}(V(0) - V^*) + V^*. \quad (4)$$

Let $\lambda_{\min}(A)$ denote the smallest eigenvalue of A . Since $A \succeq \tilde{F}^\top \tilde{F}$ and $\tilde{F}$ has full column rank, $\lambda_{\min}(A) \geq \sigma_{\min}(\tilde{F})^2$ for $\eta = 0$ and $\lambda_{\min}(A) \geq \sigma_{\min}(\tilde{F})^2 + \eta$ for $\eta > 0$. Standard bounds on matrix exponentials give

$$\|\Delta V(t)\|_F \leq e^{-\lambda_{\min}(A)t} \|\Delta V(0)\|_F. \quad (5)$$

The error satisfies

$$E(t) = \tilde{Y} - \tilde{F}V(t) = \tilde{Y} - \tilde{F}V^* - \tilde{F}\Delta V(t) =: E^* - \tilde{F}\Delta V(t),$$

where $E^* := \tilde{Y} - \tilde{F}V^*$ is the residual at the minimizer. Using $\|\tilde{F}\Delta V(t)\|_F \leq \|\tilde{F}\|_2 \|\Delta V(t)\|_F$ and (5),

$$\|E(t) - E^*\|_F \leq \|\tilde{F}\|_2 e^{-\lambda_{\min}(A)t} \|\Delta V(0)\|_F. \quad (6)$$

1.2.2 CASE $\eta = 0$

When $\eta = 0$, we have $A = \tilde{F}^\top \tilde{F}$. By assumption (A2), $\tilde{Y} = \tilde{F}V^*$ is exactly realized by the model, so $E^* = 0$, i.e.

$$\lim_{t \rightarrow \infty} E(t) = 0.$$

Equations (5) and (6) imply exponential decay:

$$\|V(t) - V^*\|_F \leq e^{-\sigma_{\min}(\tilde{F})^2 t} \|V(0) - V^*\|_F, \quad \|E(t)\|_F \leq \|\tilde{F}\|_2 e^{-\sigma_{\min}(\tilde{F})^2 t} \|V(0) - V^*\|_F.$$

We can now bound $G_{\tilde{F}}(t)$:

$$G_{\tilde{F}}(t) = E(t)V(t)^\top,$$

so

$$\|G_{\tilde{F}}(t)\|_F \leq \|E(t)\|_F \|V(t)\|_2 \leq \|E(t)\|_F (\|V^*\|_2 + \|V(t) - V^*\|_2). \quad (7)$$

Using the exponential bounds above and the fact that $\|V(t) - V^*\|_2 \leq \|V(t) - V^*\|_F$, we obtain

$$\|G_{\tilde{F}}(t)\|_F \leq C_0 e^{-\sigma_{\min}(\tilde{F})^2 t}$$

for some constant C_0 depending on $\tilde{F}$, V^* and $V(0)$ but not on t . Thus in the realizable, unregularized case, the backpropagated gradient decays exponentially to zero.

Proposition 2 (Exponential decay of $G_{\tilde{F}}$ for $\eta = 0$). *Assume (A1)–(A2) and $\eta = 0$. Then*

$$\lim_{t \rightarrow \infty} G_{\tilde{F}}(t) = 0,$$

and there exists $C_0 > 0$ such that

$$\|G_{\tilde{F}}(t)\|_F \leq C_0 e^{-\sigma_{\min}(\tilde{F})^2 t} \quad \text{for all } t \geq 0.$$

A more refined analysis using the SVD $\tilde{F} = U\Sigma W^\top$ shows that every singular direction of $G_{\tilde{F}}(t)$ is a finite linear combination of exponentials $e^{-(\sigma_i^2 + \sigma_V^2)t}$ and $e^{-\sigma_i^2 t}$, so the slowest rate in the Frobenius norm is indeed $e^{-\sigma_{\min}(\tilde{F})^2 t}$.

Proposition 1 ($G_{\tilde{F}}$ 的小时间主导项). 在假设 (A1)–(A2) 及尺度 $\alpha \ll 1$ 的小随机初始化下,

$$G_{\tilde{F}}(t) = \tilde{Y}V_0^\top + t\tilde{Y}\tilde{Y}^\top \tilde{F} + O(t\alpha + t^2)$$

在 Frobenius 范数意义下成立。特别地, 当 $\alpha \rightarrow 0$ 时,

$$\frac{G_{\tilde{F}}(t) - \tilde{Y}V_0^\top}{t} \rightarrow \tilde{Y}\tilde{Y}^\top \tilde{F}, \quad \text{且} \quad \|G_{\tilde{F}}(t)\|_F \sim t \|\tilde{Y}\tilde{Y}^\top \tilde{F}\|_F$$

对于固定的较小 t 成立, 且与 $\eta = 0$ 或 $\eta > 0$ 无关。

关于 η 作用的备注。 权重衰减参数 η 仅出现在涉及 V_0 的乘积中, 因此所有依赖于 η 的 t 阶贡献在范数上也是 $O(\alpha)$ 。所以, 主导的确定性项 $\tilde{Y}\tilde{Y}^\top \tilde{F}$ 对于 $\eta = 0$ 和 $\eta > 0$ 是相同的。

1.2 $G_{\tilde{F}}$ 的长时间衰减

我们现在分析 $G_{\tilde{F}}(t)$ 在 $t \rightarrow \infty$ 时的行为, 同样针对 $\eta = 0$ 和 $\eta > 0$ 两种情况。

1.2.1 梯度流的通解

由 (??), 梯度流是一个常系数线性常微分方程。唯一的固定点 V^* 满足

$$AV^* = B \quad \Rightarrow \quad V^* = A^{-1}B.$$

定义 $\Delta V(t) := V(t) - V^*$ 。那么

$$\frac{d}{dt}\Delta V(t) = -A\Delta V(t), \quad \Delta V(t) = e^{-At}\Delta V(0),$$

因此

$$V(t) = e^{-At}(V(0) - V^*) + V^*. \quad (4)$$

令 $\lambda_{\min}(A)$ 表示 A 的最小特征值。由于 $A \succeq \tilde{F}^\top \tilde{F}$ 且 $\tilde{F}$ 列满秩, 当 $\eta = 0$ 时 $\lambda_{\min}(A) \geq \sigma_{\min}(\tilde{F})^2$, 当 $\eta > 0$ 时 $\lambda_{\min}(A) \geq \sigma_{\min}(\tilde{F})^2 + \eta$ 。矩阵指数的标准界给出

$$\|\Delta V(t)\|_F \leq e^{-\lambda_{\min}(A)t} \|\Delta V(0)\|_F. \quad (5)$$

误差满足

$$E(t) = \tilde{Y} - \tilde{F}V(t) = \tilde{Y} - \tilde{F}V^* - \tilde{F}\Delta V(t) =: E^* - \tilde{F}\Delta V(t),$$

其中 $E^* := \tilde{Y} - \tilde{F}V^*$ 是最小值点处的残差。利用 $\|\tilde{F}\Delta V(t)\|_F \leq \|\tilde{F}\|_2 \|\Delta V(t)\|_F$ 和 (5),

$$\|E(t) - E^*\|_F \leq \|\tilde{F}\|_2 e^{-\lambda_{\min}(A)t} \|\Delta V(0)\|_F. \quad (6)$$

1.2.2 情形 $\eta = 0$

当 $\eta = 0$ 时, 我们有 $A = \tilde{F}^\top \tilde{F}$ 。根据假设 (A2), 模型恰好能实现 $\tilde{Y} = \tilde{F}V^*$, 所以 $E^* = 0$, 即

$$\lim_{t \rightarrow \infty} E(t) = 0.$$

方程 (5) 和 (6) 意味着指数衰减:

$$\|V(t) - V^*\|_F \leq e^{-\sigma_{\min}(\tilde{F})^2 t} \|V(0) - V^*\|_F, \quad \|E(t)\|_F \leq \|\tilde{F}\|_2 e^{-\sigma_{\min}(\tilde{F})^2 t} \|V(0) - V^*\|_F.$$

现在我们可以界定 $G_{\tilde{F}}(t)$:

$$G_{\tilde{F}}(t) = E(t)V(t)^\top,$$

因此

$$\|G_{\tilde{F}}(t)\|_F \leq \|E(t)\|_F \|V(t)\|_2 \leq \|E(t)\|_F (\|V^*\|_2 + \|V(t) - V^*\|_2). \quad (7)$$

1.2.3 CASE $\eta > 0$

When $\eta > 0$, the minimizer $V^* = A^{-1}\tilde{F}^\top \tilde{Y}$ is the ridge solution. In general it does *not* exactly interpolate $\tilde{Y}$, and the residual

$$E^* := \tilde{Y} - \tilde{F}V^*$$

is nonzero. Consequently the limiting backpropagated gradient

$$G_{\tilde{F}}^* := \lim_{t \rightarrow \infty} G_{\tilde{F}}(t) = E^* V^{*\top}$$

is also nonzero in general. To study the convergence, write

$$G_{\tilde{F}}(t) - G_{\tilde{F}}^* = E(t)V(t)^\top - E^*V^{*\top} = (E(t) - E^*)V(t)^\top + E^*(V(t) - V^*)^\top.$$

Using (5)–(6) and $\|AB\|_F \leq \|A\|_F \|B\|_2$, we obtain

$$\begin{aligned} \|G_{\tilde{F}}(t) - G_{\tilde{F}}^*\|_F &\leq \|E(t) - E^*\|_F \|V(t)\|_2 + \|E^*\|_F \|V(t) - V^*\|_2 \\ &\leq \left(\|\tilde{F}\|_2 \|V(0) - V^*\|_F \|V(t)\|_2 + \|E^*\|_F \|V(0) - V^*\|_F \right) e^{-\lambda_{\min}(A)t}. \end{aligned}$$

Since $\|V(t)\|_2$ is bounded (it converges to $\|V^*\|_2$), this shows exponential convergence of $G_{\tilde{F}}(t)$ to $G_{\tilde{F}}^*$. Therefore, we have the following proposition:

Proposition 3 (Exponential convergence of $G_{\tilde{F}}$ for $\eta > 0$). *Assume (A1)–(A2) and $\eta > 0$. Then*

$$\lim_{t \rightarrow \infty} G_{\tilde{F}}(t) = G_{\tilde{F}}^* := E^* V^{*\top} \neq 0 \text{ in general,}$$

and there exists $C_1 > 0$ such that

$$\|G_{\tilde{F}}(t) - G_{\tilde{F}}^*\|_F \leq C_1 e^{-\lambda_{\min}(A)t}, \quad \lambda_{\min}(A) \geq \sigma_{\min}(\tilde{F})^2 + \eta.$$

Finally, note that

$$G_{\tilde{F}}^* = E^* V^{*\top} = P_\eta \tilde{Y} V^{*\top} = \eta(\tilde{F}\tilde{F}^\top + \eta I)^{-1} \tilde{Y} \tilde{Y}^\top \tilde{F}(\tilde{F}^\top \tilde{F} + \eta I)^{-1} \quad (8)$$

where $P_\eta := I - \tilde{F}(\tilde{F}^\top \tilde{F} + \eta I)^{-1} \tilde{F}^\top = \eta(\tilde{F}\tilde{F}^\top + \eta I)^{-1}$, by Woodbury matrix formula.

Summary.

- For small t , the leading term in $G_{\tilde{F}}(t)$ is $t \tilde{Y} \tilde{Y}^\top \tilde{F}$, independent of η . All other terms (including those involving $V(0)$ and η) are lower order in the initialization scale α .
- For $\eta = 0$ and realizable $\tilde{Y} \in \text{col}(\tilde{F})$, both the error $E(t)$ and $G_{\tilde{F}}(t)$ decay exponentially to zero at rate at least $\sigma_{\min}(\tilde{F})^2$.
- For $\eta > 0$, $E(t)$ and $V(t)$ converge exponentially to (E^*, V^*) , and $G_{\tilde{F}}(t)$ converges exponentially to a nonzero limit $G_{\tilde{F}}^* = E^* V^{*\top}$.
- the total probability on distinctive tokens $q(t) := \sum_{l \in \mathcal{D}_i} p_{il}(t)$ satisfies $\dot{q} \geq q(1 - q)\Gamma$ and $q(t) \geq \gamma$ for all $t \geq T$;
- for every common token $l \in \mathcal{C}_i$,

$$\dot{p}_{il}(t) \leq -q(t)\Gamma(t)p_{il}(t), \quad p_{il}(t) \leq p_{il}(T) \exp\left(-\int_T^t q(s)\Gamma(s)ds\right) \rightarrow 0.$$

Proof. By Thm. ??, $\dot{p}_{il} = p_{il}(r_{il} - \bar{r}_i)$. Averaging over $l \in \mathcal{D}_i$ vs $l \in \mathcal{C}_i$ gives

$$\dot{q} = \sum_{l \in \mathcal{D}_i} p_{il}(r_{il} - \bar{r}_i) = \sum_{l \in \mathcal{D}_i} p_{il} r_{il} - q \bar{r}_i = (1 - q)A - q(1 - q) \underbrace{\sum_{l \in \mathcal{C}_i} \frac{p_{il}}{1 - q} r_{il}}_{\leq c(t)} \geq q(1 - q)\Gamma(t),$$

where $\Gamma(t) := d(t) - c(t)$ and we used $A \geq q d(t)$. Also, for each common l ,

$$\dot{p}_{il} = p_{il}(r_{il} - \bar{r}_i) \leq p_{il}(c(t) - [q d(t) + (1 - q)c(t)]) = -q(t)\Gamma(t)p_{il}.$$

By Lemma ??, at $t = 0$ we have $\Gamma(0) \geq \delta_0 - \varepsilon > 0$, hence $\dot{q} \geq q(1 - q)\Gamma(0)$ and $q(t)$ increases logarithmically to cross any fixed $\gamma \in (0, 1)$ in finite time T . For $t \geq T$, $q(t) \geq \gamma$ and $\Gamma(t) \geq \delta_0 - \varepsilon - (2\sqrt{2} + C)R \text{TV}_{\max}(t)$. Since $\text{TV}_{\max}(t)$ remains bounded (and typically decreases as mass moves to distinctive tokens), the integral $\int_T^t q(s)\Gamma(s)ds$ diverges, yielding $p_{il}(t) \rightarrow 0$. A fortiori, if attentions remain diffuse enough so that $\Gamma(t) \geq \delta_0 - \varepsilon - \epsilon_*$ > 0, then $p_{il}(t) \leq p_{il}(T) e^{-\gamma(\delta_0 - \varepsilon - \epsilon_*)(t - T)}$ decays exponentially. $\square$

利用上述指数界以及 $\|V(t) - V^*\|_2 \leq \|V(t) - V^*\|_F$ 这一事实，我们得到

$$\|G_{\tilde{F}}(t)\|_F \leq C_0 e^{-\sigma_{\min}(\tilde{F})^2 t}$$

其中常数 C_0 依赖于 $\tilde{F}$ 、 V^* 和 $V(0)$ ，但不依赖于 t 。因此，在可实现且无正则化的情形下，反向传播的梯度指数衰减至零。

Proposition 2 ($\eta = 0$ 时 $G_{\tilde{F}}$ 的指数衰减). 假设 (A1)–(A2) 且 $\eta = 0$ 。那么

$$\lim_{t \rightarrow \infty} G_{\tilde{F}}(t) = 0,$$

且存在 $C_0 > 0$ 使得

$$\|G_{\tilde{F}}(t)\|_F \leq C_0 e^{-\sigma_{\min}(\tilde{F})^2 t} \quad \text{对所有 } t \geq 0.$$

使用奇异值分解 $\tilde{F} = U\Sigma W^\top$ 的更精细分析表明， $G_{\tilde{F}}(t)$ 的每个奇异方向都是指数函数 $e^{-(\sigma_i^2 + \sigma_j^2)t}$ 和 $e^{-\sigma_i^2 t}$ 的有限线性组合，因此 Frobenius 范数下的最慢衰减速率确实是 $e^{-\sigma_{\min}(\tilde{F})^2 t}$ 。

1.2.3 情形 $\eta > 0$

当 $\eta > 0$ 时，最小化器 $V^* = A^{-1}\tilde{F}^\top \tilde{Y}$ 是岭回归解。通常它并不精确插值 $\tilde{Y}$ ，且残差

$$E^* := \tilde{Y} - \tilde{F}V^*$$

非零。因此，极限反向传播梯度

$$G_{\tilde{F}}^* := \lim_{t \rightarrow \infty} G_{\tilde{F}}(t) = E^* V^{*\top}$$

通常也非零。为研究收敛性，记

$$G_{\tilde{F}}(t) - G_{\tilde{F}}^* = E(t)V(t)^\top - E^*V^{*\top} = (E(t) - E^*)V(t)^\top + E^*(V(t) - V^*)^\top.$$

利用 (5)–(6) 及 $\|AB\|_F \leq \|A\|_F \|B\|_2$ ，我们得到

$$\begin{aligned} \|G_{\tilde{F}}(t) - G_{\tilde{F}}^*\|_F &\leq \|E(t) - E^*\|_F \|V(t)\|_2 + \|E^*\|_F \|V(t) - V^*\|_2 \\ &\leq \left(\|\tilde{F}\|_2 \|V(0) - V^*\|_F \|V(t)\|_2 + \|E^*\|_F \|V(0) - V^*\|_F \right) e^{-\lambda_{\min}(A)t}. \end{aligned}$$

由于 $\|V(t)\|_2$ 有界（它收敛到 $\|V^*\|_2$ ），这表明 $G_{\tilde{F}}(t)$ 以指数速度收敛到 $G_{\tilde{F}}^*$ 。因此，我们有以下命题：

Proposition 3 ($\eta > 0$ 时 $G_{\tilde{F}}$ 的指数收敛性). 假设 (A1)–(A2) 且 $\eta > 0$ 。则

$$\lim_{t \rightarrow \infty} G_{\tilde{F}}(t) = G_{\tilde{F}}^* := E^* V^{*\top} \neq 0 \text{ 通常成立,}$$

且存在 $C_1 > 0$ 使得

$$\|G_{\tilde{F}}(t) - G_{\tilde{F}}^*\|_F \leq C_1 e^{-\lambda_{\min}(A)t}, \quad \lambda_{\min}(A) \geq \sigma_{\min}(\tilde{F})^2 + \eta.$$

最后，注意

$$G_{\tilde{F}}^* = E^* V^{*\top} = P_\eta \tilde{Y} V^{*\top} = \eta(\tilde{F}\tilde{F}^\top + \eta I)^{-1} \tilde{Y} \tilde{Y}^\top \tilde{F}(\tilde{F}^\top \tilde{F} + \eta I)^{-1} \quad (8)$$

其中 $P_\eta := I - \tilde{F}(\tilde{F}^\top \tilde{F} + \eta I)^{-1} \tilde{F}^\top = \eta(\tilde{F}\tilde{F}^\top + \eta I)^{-1}$ ，由 Woodbury 矩阵公式可得。

总结。

- 对于小的 t ， $G_{\tilde{F}}(t)$ 中的主导项是 $t \tilde{Y} \tilde{Y}^\top \tilde{F}$ ，与 η 无关。所有其他项（包括涉及 $V(0)$ 和 η 的项）在初始化尺度 α 下都是低阶的。

- 对于 $\eta = 0$ 且 $\tilde{Y} \in \text{col}(\tilde{F})$ 可实现的情形, 误差 $E(t)$ 和 $G_{\tilde{F}}(t)$ 均以至少 $\sigma_{\min}(\tilde{F})^2$ 的速率指数衰减到零。
- 对于 $\eta > 0$, $E(t)$ 和 $V(t)$ 指数收敛到 (E^*, V^*) , 且 $G_{\tilde{F}}(t)$ 指数收敛到一个非零极限 $G_{\tilde{F}}^* = E^* V^{*\top}$ 。
- 在独特标记上的总概率 $q(t) := \sum_{l \in \mathcal{D}_i} p_{il}(t)$ 满足 $\dot{q} \geq q(1-q)\Gamma$ 且对所有 $t \geq T$ 有 $q(t) \geq \gamma$;
- 对每个常见标记 $l \in \mathcal{C}_i$,

$$\dot{p}_{il}(t) \leq -q(t)\Gamma(t)p_{il}(t), \quad p_{il}(t) \leq p_{il}(T) \exp\left(-\int_T^t q(s)\Gamma(s)ds\right) \rightarrow 0.$$

证明. 由定理 ??, $\dot{p}_{il} = p_{il}(r_{il} - \bar{r}_i)$ 。对 $l \in \mathcal{D}_i$ 与 $l \in \mathcal{C}_i$ 分别取平均得

$$\dot{q} = \sum_{l \in \mathcal{D}_i} p_{il}(r_{il} - \bar{r}_i) = \underbrace{\sum_{l \in \mathcal{D}_i} p_{il}r_{il}}_A - q\bar{r}_i = (1-q)A - q(1-q) \underbrace{\sum_{l \in \mathcal{C}_i} \frac{p_{il}}{1-q}r_{il}}_{\leq c(t)} \geq q(1-q)\Gamma(t),$$

其中 $\Gamma(t) := d(t) - c(t)$, 且我们使用了 $A \geq qd(t)$ 。此外, 对每个常见标记 l ,

$$\dot{p}_{il} = p_{il}(r_{il} - \bar{r}_i) \leq p_{il}(c(t) - [qd(t) + (1-q)c(t)]) = -q(t)\Gamma(t)p_{il}.$$

由引理 ??, 在 $t = 0$ 时我们有 $\Gamma(0) \geq \delta_0 - \varepsilon > 0$, 因此 $\dot{q} \geq q(1-q)\Gamma(0)$ 且 $q(t)$ 以逻辑增长方式上升, 在有限时间 T 内穿过任意固定的 $\gamma \in (0, 1)$ 。对于 $t \geq T$, 有 $q(t) \geq \gamma$ 且 $\Gamma(t) \geq \delta_0 - \varepsilon - (2\sqrt{2} + C)R\text{TV}_{\max}(t)$ 。由于 $\text{TV}_{\max}(t)$ 保持有界 (且通常随着质量转移到独特标记而减小), 积分 $\int_T^t q(s)\Gamma(s)ds$ 发散, 从而得到 $p_{il}(t) \rightarrow 0$ 。更进一步, 如果注意力保持足够分散使得 $\Gamma(t) \geq \delta_0 - \varepsilon - \epsilon_* > 0$, 那么 $p_{il}(t) \leq p_{il}(T)e^{-\gamma(\delta_0 - \varepsilon - \epsilon_*)(t-T)}$ 呈指数衰减。□